

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (3 points) Find

$$\begin{aligned} & \lim_{x \rightarrow -\infty} \frac{2 - e^x}{2 + e^x} \\ &= \frac{2 - 1}{2 + 1} \quad (\text{Since } \lim_{x \rightarrow -\infty} e^x = 1) \\ &= \frac{1}{3} \end{aligned}$$

Problem 2. (4 points) Find values of x , if any, at which $f(x) = \frac{x-3}{|x|-3}$ is not continuous, and determine whether each such value is a removable discontinuity.

Solution: f is discontinuous at points $|x|-3=0$

$$\Rightarrow x=3 \text{ or } -3$$

i) When $x=3$,

$$\lim_{x \rightarrow 3} \frac{x-3}{|x|-3} = \lim_{x \rightarrow 3} \frac{x-3}{x-3} = 1$$

so it is a removable discontinuity.

ii) When $x=-3$,

$$\lim_{x \rightarrow -3} \frac{x-3}{|x|-3} = \lim_{x \rightarrow -3} \frac{x-3}{-x-3} = \lim_{x \rightarrow -3} \frac{-3-3}{-(-3)-3} = \lim_{x \rightarrow -3} \frac{6}{0}$$

does not exist.

so it is not a removable discontinuity.

Problem 3. (3 points) Find

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos^2 x} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} = \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^2 \\ &= \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} \right)^2 \\ &= 1^2 = 1 \end{aligned}$$